Control of an open-chain manipulator

- Ch 4 §3.2 derived the equations of motion of an open-chain manipulator

$$M(\theta)\ddot{\theta}+C(\theta,\dot{\theta})\dot{\theta}+N(\theta,\dot{\theta})=\tau \qquad (4.24)(4.47)$$

- Nonlinear ODE in time; can be solved for given $\tau(t)$ to determine trajectory $\theta(t)$.
- This is a mathematical model of a physical plant.
- The model can be used in building a controller for the plant.
- In simulation the plant is represented with the model.
- Therefore in simulation one typically have the model twice: to represent the plant and as part of the controller.

Control of an open-chain manipulator

When in the beginning of §5.1 (p. 190) MLS writes

$$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + N(\theta, \dot{\theta}) = \tau \qquad (4.47)$$

and then that "we may solve our problem by choosing

$$\tau = M(\theta_d)\ddot{\theta}_d + C(\theta_d, \dot{\theta}_d)\dot{\theta}_d + N(\theta_d, \dot{\theta}_d)\text{"} \ ,$$

one should really think of the 1st equation as the plant and the second as the controller (or control law).

θ_d, at least twice differentiable, is the joint trajectory we wish to track.

Open loop control law, not robust; feedback can improve this.

§5.2 Computed torque (p. 191)

Plant: $$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + N(\theta, \dot{\theta}) = \tau \qquad (4.47)$$

Control law: $$\tau = M(\theta_d)\ddot{\theta}_d + C(\theta_d, \dot{\theta}_d)\dot{\theta}_d + N(\theta_d, \dot{\theta}_d)$$

In §5.2 MLS suggest an alternative control law:

$$\tau = M(\theta)\left(\ddot{\theta}_d - K_v\dot{e} - K_p e\right) + C(\theta, \dot{\theta})\dot{\theta} + N(\theta, \dot{\theta})\,, \qquad (4.49)$$

where $e = \theta - \theta_d$ and K_v and K_p are constant feedback gain matrices.

Called the *computed torque control law*

§5.2 Computed torque (p. 191)

Plant: $$M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + N(\theta, \dot{\theta}) = \tau \qquad (4.47)$$

CL (4.49): $$\tau = M(\theta)\left(\ddot{\theta}_d - K_v\dot{e} - K_p e\right) + C(\theta, \dot{\theta})\dot{\theta} + N(\theta, \dot{\theta})$$

Substituting the control law into eq. (4.47):

$$M(\theta)\left(\ddot{e} + K_v\dot{e} + K_p e\right) = 0$$

$$\ddot{e} + K_v\dot{e} + K_p e = 0 \qquad (4.50)$$

§5.2 Computed torque (p. 191)

$$\ddot{e} + K_v \dot{e} + K_p e = 0 \tag{4.50}$$

$$\frac{d}{dt}\begin{bmatrix} e \\ \dot{e} \end{bmatrix} = \begin{bmatrix} 0 & I \\ -K_p & -K_v \end{bmatrix} \begin{bmatrix} e \\ \dot{e} \end{bmatrix}$$

Let us define

$$\mathcal{A} = \begin{bmatrix} 0 & I \\ -K_p & -K_v \end{bmatrix} .$$

We can place the eigenvalues of $\mathcal{A}$ anywhere we like by **first of all** choosing the gain matrices K_p and K_v to be positive definite diagonal matrices.

$$\mathcal{A}^T = \begin{bmatrix} 0 & -K_p^T \\ I & -K_v^T \end{bmatrix} \quad \text{and} \quad |\mathcal{A}| = \left|\mathcal{A}^T\right| .$$

§5.2 Computed torque (p. 191)

$$\mathcal{A}^T = \begin{bmatrix} 0 & -K_p^T \\ I & -K_v^T \end{bmatrix} \quad \text{and} \quad |\mathcal{A}| = \left|\mathcal{A}^T\right| \ .$$

To calculate an eigenvalues λ of $\mathcal{A}^T$one needs to calculate the roots of the characteristic equation

$$\left|\lambda I_{2n} - \mathcal{A}^T\right| = 0$$

$$\left|\lambda I_{2n} - \mathcal{A}^T\right| = \begin{vmatrix} \lambda I_n & K_p^T \\ -I_n & \lambda I_n + K_v^T \end{vmatrix} = \left|\lambda^2 I_n + \lambda K_v^T + K_p^T\right| = 0$$

§5.2 Computed torque (p. 191)

$$\left|\lambda I_{2n} - \mathcal{A}^T\right| = \begin{vmatrix} \lambda I_n & K_p^T \\ -I_n & \lambda I_n + K_v^T \end{vmatrix} = \left|\lambda^2 I_n + \lambda K_v^T + K_p^T\right| = 0$$

But since K_v and K_p are chosen to be diagonal, the determinant calculation reduces to

$$\prod_{i=1}^{n} \left(\lambda_i^2 + \lambda_i K_{v\,ii} + K_{p_{ii}}\right) = 0\ ,$$

which allows us to construct each eigenvalue to our liking by choosing the gains $K_{v\,ii}$ and $K_{p_{ii}}$.

§5.2 Computed torque (p. 191)

Control law:

$$\tau = M(\theta)\left(\ddot{\theta}_d - K_v\dot{e} - K_p e\right) + C(\theta,\dot{\theta})\dot{\theta} + N(\theta,\dot{\theta}) \quad (4.49)$$

$$\begin{aligned}\tau &= M(\theta)\ddot{\theta}_d + C(\theta,\dot{\theta})\dot{\theta} + N(\theta,\dot{\theta}) + M(\theta)(-K_v\dot{e} - K_p e) \\ &= \tau_{ff} + \tau_{fb}\end{aligned}$$

where

$$\tau_{ff} = M(\theta)\ddot{\theta}_d + C(\theta,\dot{\theta})\dot{\theta} + N(\theta,\dot{\theta}) \qquad \text{feedforward component}$$

$$\tau_{fb} = M(\theta)(-K_v\dot{e} - K_p e) \qquad \text{feedback component}$$

Proposition 4.8 p. 192 shows that the computed torque control law results in exponential trajectory tracking if $K_p, K_v \in \mathbb{R}^{n\times n}$ are positive definite symmetric.

Proof: Proposition 4.8

$$\ddot{e} + K_v \dot{e} + K_p e = 0 \qquad (4.50)$$

$$\frac{d}{dt}\begin{bmatrix} e \\ \dot{e} \end{bmatrix} = \begin{bmatrix} 0 & I \\ -K_p & -K_v \end{bmatrix}\begin{bmatrix} e \\ \dot{e} \end{bmatrix} = \mathcal{A}\begin{bmatrix} e \\ \dot{e} \end{bmatrix}$$

Let $\lambda \in \mathbb{C}$ be an eigenvalue of $\mathcal{A}$, with corresponding eigenvector $v = (v_1, v_2) \in \mathbb{C}^{2n}, v \neq 0$.

$$\lambda \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 & I \\ -K_p & -K_v \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_2 \\ -K_p v_1 - K_v v_2 \end{bmatrix}$$

$\lambda = 0$ cannot be an eigenvalue and neither v_1 nor v_2 can be 0.

Proof: Proposition 4.8

$$\lambda \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 & I \\ -K_p & -K_v \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_2 \\ -K_p v_1 - K_v v_2 \end{bmatrix}$$

We can assume $||v_1|| = 1 \Rightarrow ||v_1||^2 = v_1^* v_1 = 1$.

$$\begin{aligned} \lambda^2 &= \lambda^2 v_1^* v_1 = v_1^* \lambda^2 v_1 = v_1^* \lambda v_2 = v_1^* (-K_p v_1 - K_v v_2) \\ &= -v_1^* K_p v_1 - \lambda v_1^* K_v v_1 = -\beta - \alpha\lambda \end{aligned}$$

with $\alpha > 0$ and $\beta > 0$

$$\lambda^2 + \alpha\lambda + \beta = 0$$

which implies real$(\lambda_i) < 0 \; \forall \; i$ QED

For a square matrix that has four equal size square submatrices A, B, C and D:

$$\left[\begin{array}{cc} A & B \\ C & D \end{array}\right]$$

and where C and D communte, that is, $CD = DC$, it can be shown that the determinant of the matrix is $|AD - BC|$ (see, under "Block matrices", Wikipedia description).